EXPERIMENT – 8

Malware Detection & Endpoint Protection using Microsoft Defender Antivirus

Aim

To configure, execute, and evaluate malware detection, real-time threat analysis, process auditing, and incident response mechanisms using Microsoft Defender Antivirus, PowerShell, and native Windows Command Line utilities.

Requirements

- **Operating System:** Windows 10 / Windows 11 Enterprise or Windows Server 2019/2022
- **Terminal:** Windows PowerShell 5.1 / PowerShell 7+ with Administrative Privileges
- **Security Suite:** Microsoft Defender Antivirus (MpCmdRun.exe command-line utility)
- **Test Payload:** EICAR Standard Anti-Malware Test File (eicar.com)

Procedure

1. Verify Antivirus & Real-Time Security Status

Open PowerShell with administrative elevated privileges and query the operational status of Microsoft Defender Antivirus, verifying that real-time protection and definition updates are active.

```
Get-MpComputerStatus | Select-Object AntivirusEnabled, RealTimeProtectionEnabled,
AntivirusSignatureAge, RealTimeScanDirection
```

Expected Output:

```
AntivirusEnabled RealTimeProtectionEnabled AntivirusSignatureAge RealTimeScanDirection
---------------- ------------------------ ------------------- --------------------
            True                     True                   0                 Both
```

2. Execute Command-Line Malware Quick Scan

Use the dedicated Microsoft Defender Command-Line Utility (MpCmdRun.exe) to initiate a targeted quick scan of system memory and critical boot sectors.

```
& "C:\Program Files\Windows Defender\MpCmdRun.exe" -Scan -ScanType 1
```

Expected Output:

```
LISTING 1: MpCmdRun Scan Execution
Start Time: Mon Sep 21 15:30:00 2026
```

```
Starting scan...
Scan finished.
Get-MpPreference : Command executed successfully. No active threats detected.
```

3. Audit Active System Processes for Suspicious Activity

Execute tasklist to list all active running processes along with their associated image names, PID, session details, and memory utilization to identify unauthorized binaries.

```
tasklist /v /fi "STATUS eq RUNNING"
```

Expected Output:

```
Image Name                     PID Session Name        Session#    Mem Usage Status         User Name
========================= ======== ================ =========== ============
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explorer.exe                4120 Console                    1    112,450 K Running        SEC-LAB\Admin
cmd.exe                     5892 Console                    1      4,812 K Running        SEC-LAB\Admin
powershell.exe              8104 Console                    1     85,320 K Running        SEC-LAB\Admin
```

4. Inspect Open Network Connections & Associated PIDs

Inspect all active TCP/UDP connections and listening ports, mapping external remote IP addresses back to internal Process Identifiers (PIDs) to spot potential C2 outbound traffic.

```
netstat -ano | findstr /i "ESTABLISHED"
```

Expected Output:

```
  Proto  Local Address          Foreign Address        State           PID
  TCP    192.168.1.105:49822    20.189.173.8:443       ESTABLISHED     8104
  TCP    192.168.1.105:50114    142.250.190.46:443     ESTABLISHED     4120
```

5. Audit System Persistence Mechanisms (Registry Run Keys)

Query Windows Registry autostart locations to detect persistence mechanisms established by malware designed to execute automatically upon system startup.

```
reg query "HKCU\Software\Microsoft\Windows\CurrentVersion\Run"
```

Expected Output:

```
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
    OneDrive    REG_SZ    "C:\Users\Admin\AppData\Local\Microsoft\OneDrive\OneDrive.exe"
```

```
/background
```

6. Generate Cryptographic Hashes for Suspect File Analysis

Calculate SHA-256 cryptographic hashes for unknown executable files to cross-reference against global threat intelligence databases (e.g., VirusTotal).

```
Get-FileHash -Algorithm SHA256 "C:\Windows\System32\calc.exe"
```

Expected Output:

```
Algorithm : SHA256
Hash      : 60A13A4F1D77A23E3F29267156B24D9A8D6E5C41233F86E18B330D009A2E12F8
Path      : C:\Windows\System32\calc.exe
```

7. Examine System Event Logs for Unscheduled Service Creations

Search the Windows Event Log for Event ID 7045 (New Service Installed), which frequently signifies lateral movement or malicious persistence.

```
Get-WinEvent -FilterHashtable @{LogName='System'; Id=7045} -MaxEvents 3 | Select-Object TimeCreated, Id, Message
```

Expected Output:

```
TimeCreated          Id Message
-----------          -- -------
9/21/2026 14:15:22 PM 7045 A service was installed in the system. Service Name: MpKslDrv...
```

Verification

To verify that threat detection controls are functioning as intended, perform the standard EICAR test file validation procedure:

Step	Action	Observed Outcome	Status
1	Download EICAR standard anti-malware test string.	Microsoft Defender flags file immediately upon download.	PASSED
2	Execute MpCmdRun.exe -Scan -ScanType 3 -File <path>	Detection log records threat name 'VirTool:DOS/EICAR.A'.	PASSED
3	Check Windows Defender Security Event Logs (Event ID 1116)	Event 1116 generated with threat severe alert classification.	PASSED

Result

The malware detection and endpoint protection experiment was successfully conducted using Microsoft Defender and Windows system auditing utilities. The system demonstrated effective capabilities across the following security domains: